Capstone Section 3 SQL

Setup

```
mysql> SELECT * FROM users;
+---------+---------------+-----------------------+-------------+
| user_id | name          | email                 | password    |
+---------+---------------+-----------------------+-------------+
|       1 | Alice Johnson | alice.new@example.com | alice123    |
|       2 | Bob Smith     | bob@example.com       | bob123      |
|       3 | Charlie Lee   | charlie@example.com   | charlie123  |
+---------+---------------+-----------------------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM courses;
+-----------+---------------------+-----------------------------------------+
| course_id | course_name         | description                             |
+-----------+---------------------+-----------------------------------------+
|         1 | HTML Basics         | Introduction to HTML and web structure. |
|         2 | CSS Design          | Learn how to style websites using CSS.  |
|         3 | MySQL for Beginners | Basic concepts of relational databases. |
+-----------+---------------------+-----------------------------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM enrollments;
+---------------+---------+-----------+-----------------+
| enrollment_id | user_id | course_id | enrollment_date |
+---------------+---------+-----------+-----------------+
|             1 |       1 |         1 | 2024-01-10      |
|             3 |       2 |         2 | 2024-02-15      |
|             4 |       3 |         1 | 2024-03-01      |
+---------------+---------+-----------+-----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM assessments;
+---------------+-----------+-----------------+-----------+
| assessment_id | course_id | title           | max_score |
+---------------+-----------+-----------------+-----------+
|             1 |         1 | HTML Quiz 1     |       100 |
|             2 |         2 | CSS Midterm     |        80 |
|             3 |         3 | MySQL Final Test|        90 |
+---------------+-----------+-----------------+-----------+
3 rows in set (0.00 sec)
```

1. Create a new table named instructors:
    - instructor_id (Primary Key, auto-increment)
    - name (varchar)
    - email (unique)

```
mysql>
mysql> CREATE TABLE instructors ( instructor_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100), email VARCHAR(100) UNIQUE );
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO instructors (name, email) VALUES ('Dr. Sarah Johnson', 'sarah.johnson@lms.edu'), ('Prof. Michael Chen', 'michael.chen@lms.e
du'), ('Ms. Emily Rodriguez', 'emily.rodriguez@lms.edu');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * from instructors;
+---------------+--------------------+--------------------------+
| instructor_id | name               | email                    |
+---------------+--------------------+--------------------------+
|             1 | Dr. Sarah Johnson  | sarah.johnson@lms.edu    |
|             2 | Prof. Michael Chen | michael.chen@lms.edu     |
|             3 | Ms. Emily Rodriguez | emily.rodriguez@lms.edu |
+---------------+--------------------+--------------------------+
3 rows in set (0.00 sec)

mysql>
```

2. Enroll new user Daniel Rose (daniel@lms.com, password: daniel123) in "CSS Design"
    - Add this user to the users table
    - Enroll him in the "CSS Design" course with today's date
    - Write a query to display all users enrolled in "CSS Design"

```
mysql>
mysql> INSERT INTO users (name, email, password) VALUES ('Daniel Rose', 'daniel@lms.com', 'daniel123');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT user_id FROM users WHERE name = 'Daniel Rose';
+---------+
| user_id |
+---------+
|       4 |
+---------+
1 row in set (0.00 sec)

mysql> SELECT course_id FROM courses WHERE course_name = 'CSS Design';
+-----------+
| course_id |
+-----------+
|         2 |
+-----------+
1 row in set (0.00 sec)

mysql>
mysql> INSERT INTO enrollments (user_id, course_id, enrollment_date) VALUES (4, 2, CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql> SELECT u.name, c.course_name FROM enrollments e JOIN users u ON e.user_id = u.user_id JOIN courses c ON e.course_id = c.course_id WHERE
 c.course_name = 'CSS Design';
+-------------+-------------+
| name        | course_name |
+-------------+-------------+
| Bob Smith   | CSS Design  |
| Daniel Rose | CSS Design  |
+-------------+-------------+
2 rows in set (0.00 sec)

mysql>
```